

Inventory Forecasting and Par Level Recommendation

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Project: AI Innovations — Hotel Bar Inventory Optimization & Par Recommendation

Date: 11th November 2025

1. Core Business Problem

Hotels and bars often face the problem of having **too much stock of slow-moving brands** and **not enough of fast-selling ones**.

When a popular brand runs out, it directly affects sales and the guest experience.

On the other hand, overstocking drinks that move slowly increases cost and storage space.

The goal of this project was to build a **forecasting-based system** that helps decide *how much to keep (Par Level)* and *when to reorder (ROP)* for each bar and brand.

A system like this helps managers keep enough stock to serve guests without wasting money on overstock.

2. Assumptions Made

While working on the dataset, I made some practical assumptions to make the model realistic and simple to maintain:

- **Lead time (L)** – 3 days for suppliers to deliver new stock.
- **Service level (z)** – 95%, which is a good balance between cost and availability.
- **Demand source** – daily consumption data is considered actual demand.
- **Frequency** – data is aggregated by *bar name* and *brand name* for each date.
- **Zero values** – zero consumption days mean no sale happened, not a missing entry.

These assumptions were chosen because they fit typical hotel operations and make the forecast easy to interpret.

3. Model Used and Reason

I used a **LightGBM regression model** as the main forecasting engine for regular SKUs, and **Croston's method** for items with very irregular or intermittent sales.

Why LightGBM:

- Handles multiple SKUs and time series together efficiently.
- Works well with tabular features such as day of week, lags, and rolling averages.
- Fast to train and tune even on a laptop.

Why not ARIMA or Prophet:

They work fine for one time series but not when you have hundreds of bar-brand combinations. LightGBM can learn global patterns across all bars together, which improves consistency.

So the final model setup was:

- Create lag and rolling window features for each SKU.
- Train one global model across all bars.
- Predict daily demand (μ).
- Use training residuals to estimate uncertainty (σ) and compute Safety Stock, ROP, and Par.

4. System Performance and Improvements

The model gave good results during testing. The **mean absolute error (MAE)** was around **100–200 ml per day** for most SKUs, which is accurate enough for operational use.

In the simulation comparing the new policy vs a simple average-based baseline:

- Average inventory reduced by about **15–20%**,
- Fill rate stayed roughly the same (~95%),
- Stockout days reduced slightly for high-volume SKUs.

Possible improvements:

- Include event or holiday data (like weekends or festivals).
- Add supplier reliability to vary lead time dynamically.
- Introduce automatic alerts or dashboards for managers.
- Handle case pack rounding (bottle/carton multiples).

5. How It Would Work in a Real Hotel

In a real setup:

1. Each bar uploads daily stock and consumption data.
2. The system forecasts demand for the next week.
3. It calculates:
 - **Safety Stock** = $z \times \sigma \times \sqrt{L}$
 - **ROP** = $\mu \times L + \text{Safety Stock}$
 - **Par** = $\mu \times (L + 1) + \text{Safety Stock}$
4. These numbers are shown in a dashboard or emailed to bar managers.
5. When the current stock drops below ROP, an order is automatically suggested.

This process ensures popular brands never go out of stock while keeping storage cost under control.

Optional: Scaling and Monitoring

If the hotel chain grows or adds new locations, the system can scale easily. The main challenges would be data delays, new SKU naming, or inconsistent input.

For production monitoring, I would track:

- Forecast error (MAE or MAPE)
- Fill rate per SKU

- Average inventory days
- Lead time variations

With these metrics and retraining every few months, the system would stay accurate and reliable for long-term use.